Aryaman Raj Jaiswal

+91 9136816530 | aryamanj250@gmail.com | linkedin.com/in/aryamanrajjaiswal
github.com/aryamanraj2 | aryamanj.in

Education

Netaji Subhas University of Technology — Dwarka, Delhi
Bachelor of Technology in Electronics and Communications — *Aug. 2024 – Present*

Experience

Bharti Airtel Limited — June 2026 – Aug. 2026
Software Engineer Intern — *Gurugram, India*

- Re-architected an internal **OpenStack** platform tracking live and historical state across a distributed VM fleet; cut dashboard query latency by **30%** via **database migration**, **composite indexing**, and **cursor-based pagination**.
- Designed a guardrailed natural-language interface over the VM database using local **Ollama** LLMs (**LangChain**), **NeMo Guardrails** input/output rails, and a read-only **text-to-SQL** agent, plus an **AI surge-analysis** layer that flags anomalous provisioning spikes and summarizes root causes.

SmartOut Media (Canada) — Jan. 2026 – Present
iOS Developer (Contract) — *Remote*

- Built the **SmartOut** iOS app from the ground up in **Swift/SwiftUI** (**50K+ installs**) — an offline-first geospatial platform for Ontario hunting and fishing regulations — over a bundled **SQLite** dataset and a migratable **GRDB** store, instrumented with **GA4** analytics and **Crashlytics**.
- Engineered the map layer with **Google Maps SDK** and **GEOSwift**: **456 regulatory polygons** with MULTIPOLYGON and interior-hole support, distance-based marker clustering, polygon hit-testing, and deep links that restore the exact zone, tab, and regulation row across app launches.

Examination Cell, NSUT — May 2025 – Aug. 2025
Software Developer Intern — *Delhi, India*

- Architected a **Next.js/Flask** automation platform serving **10,000+ students**, cutting manual data entry by **70%** with **Pandas**-driven **constraint solvers** for invigilator assignment and room allocation.
- Shipped **NSUTrack**, native **Android and iOS** clients centralizing schedules, attendance, and society notices for **1,500+ students**, with an offline-first cache refreshed by **WorkManager** sync and **FCM** push.

Achievements

- **Apple Swift Student Challenge Winner (2026):** Selected among the top global student developers for **Saldo**, an on-device personal finance iOS app (**SwiftUI**, **Vision**, **Core ML**, **on-device LLM**) that runs receipt parsing and spend intelligence entirely offline.
- **Winner, MLH Brainwave 2.0 (DTU):** Led Team Aquacult to **1st place over 200+ teams**, sweeping **Best Overall Project**, **Best Use of Gemini**, and **Best UI/UX Runner-Up**.
- **2x Winner, HackonHills (2024, 2025):** Back-to-back **1st-place** finishes among **125+ teams**, building **Aqua-Sense** (AI/IoT) and **HimYatra** (Mobile/Web) end to end.

Projects

Saldo | *Kotlin, Coroutines/Flow, SQLite/Room, JUnit, Swift/SwiftUI* — 2026

- Rebuilt an unstructured **bank-SMS transaction parser** as a **staged rule engine** over **9 Indian bank/UPI templates**, fixing **13 correctness defects** — including quoted balances read as transaction amounts and **Rs. 0** rows that corrupted every dashboard aggregate.
- Wrote a **41-case golden corpus** with a regression test per defect, growing the suite from **0 to 57 passing tests**.
- Designed **idempotent ingest** keyed on a **content hash** with a **watermark advancing only after commit**, making replays safe and replacing a full-inbox rescan that ran twice per launch on the main thread.

Wandr | *Swift, Cloudflare Durable Objects, Supabase Realtime, WebSockets, Google Places/Routes API, Foundation Models* — 2026

- Built a **real-time multiplayer lobby** on **Cloudflare Durable Objects** and **Supabase Realtime**: join-code/QR sessions, **WebSocket** presence and broadcast tallies at **sub-second consistency**, **quorum locking** under host authority, and pseudonymous device-scoped IDs needing no guest accounts.
- Engineered a **seven-stage planning coordinator** (extract → normalize → research → resolve → curate → validate → schedule) where **Foundation Models** only rank pre-vetted candidates and **deterministic validation is the sole authority** permitted to mint a plan — eliminating hallucinated output by construction.
- Designed a **constraint-ladder resolver** that relaxes the least critical rule first (setting → budget → time → stops) while **never relaxing dietary or accessibility**, so planning cannot dead-end; built **hybrid retrieval** over live **Google Places + Routes** APIs behind a swappable `VenueResearching` protocol.

Technical Skills

Languages: Java, Python, Go, SQL (PostgreSQL), JavaScript, Kotlin, Swift, C/C++
Backend: REST API design, WebSockets, Flask, FastAPI, Node.js, Next.js, PostgreSQL, SQLite, Docker, OpenStack
CS Fundamentals: Data Structures & Algorithms, System Design, Distributed Systems, Concurrency, Database Design, OOP
AI / ML: LangChain, Ollama, NeMo Guardrails, RAG, Text-to-SQL, Random Forest, Foundation Models (on-device), Gemini
Cloud & Mobile: Cloudflare Durable Objects, Supabase, Firebase, Git, SwiftUI, Jetpack Compose